

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

C02: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 6

Total Marks:30

Annexure A

Scenario-Based

Code of Conduct:

*I DARSHINIE KARANI.V.M (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Scenario-Based

1. Use Case Diagram Scenario

A **Smart Cinema Ticket Booking System** allows customers to search movies, select seats, book tickets, cancel bookings, and make online payments. Theater managers update movie schedules and seat availability, while administrators manage users and system settings. The interactions between actors and system functions are not clearly defined.

Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.

(5 Marks)

2. Class Diagram Scenario

A **Smart College Management System** consists of entities such as Student, Faculty, Department, Course, Attendance, and Examination. The existing design does not clearly define class attributes, operations, or relationships among these entities.

Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.

(5 Marks)

3. Sequence Diagram Scenario

In an **Online Railway Reservation System**, a passenger searches for trains, selects a seat, books a ticket, makes payment, and receives an e-ticket. The communication among the passenger, reservation system, payment gateway, and database is incomplete.

Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.

(5 Marks)

4. Activity Diagram Scenario

A **Smart Insurance Claim Processing System** allows customers to submit insurance claims, upload supporting documents, verify claim details, approve or reject claims, and receive compensation. The workflow does not clearly represent decision-making or concurrent activities.

Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.

(5 Marks)

5. State Diagram Scenario

A **Smart Washing Machine System** operates through states such as Idle, Water Filling, Washing, Rinsing, Spinning, Paused, and Completed. The transitions between these operational states are not clearly specified.

Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the washing machine's behavior.

(5 Marks)

6. Deployment Diagram Scenario

A **Smart Energy Management System** uses smart electricity meters, edge gateways, cloud servers, utility monitoring dashboards, and mobile applications to monitor energy consumption and control power distribution. The deployment architecture does not clearly illustrate hardware nodes, deployed software artifacts, or communication links.

Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.

(5 Marks)

RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SCENARIO BASED QUESTIONS

1. A Smart Cinema Ticket Booking System allows customers to search for movies, select seats, book tickets, cancel bookings, and make online payments. Theater managers update movie schedules and seat availability, while administrators manage users and system settings.

Actors

1. Customer
2. Theater Manager
3. Administrator
4. Payment Gateway (External System)

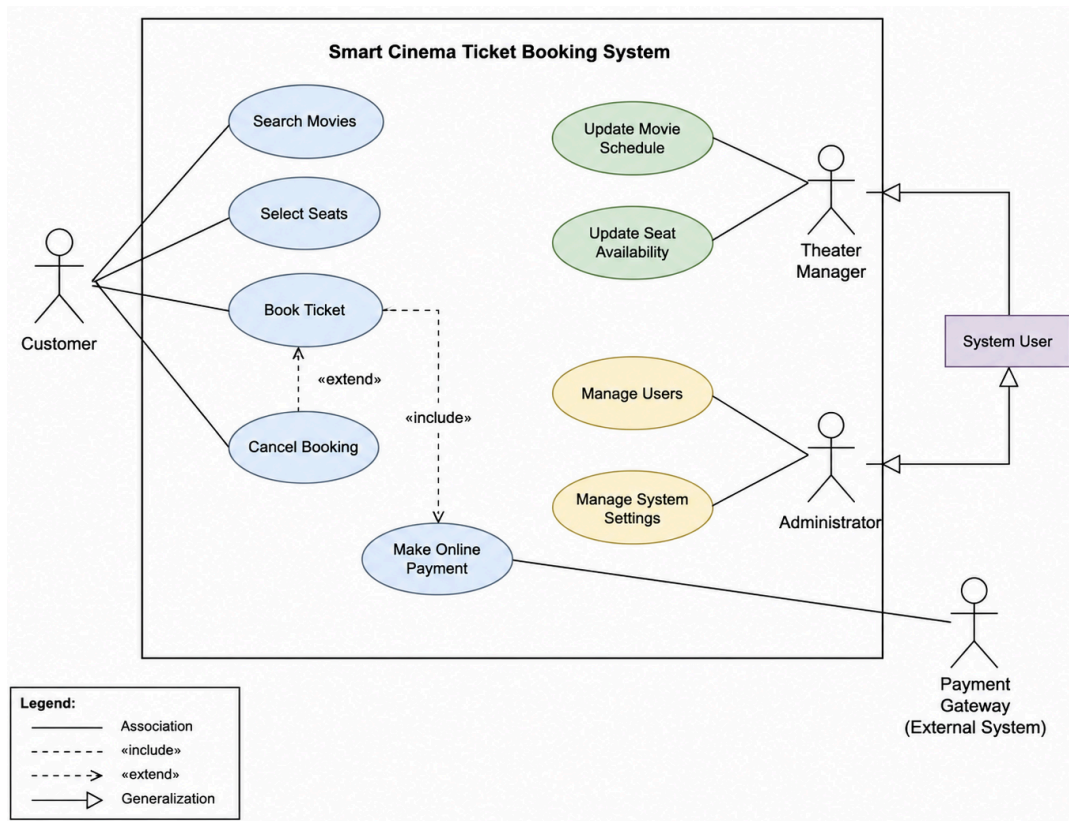
Use Cases

- Search Movies
- Select Seats
- Book Ticket
- Make Online Payment
- Cancel Booking
- Update Movie Schedule
- Update Seat Availability
- Manage Users
- Manage System Settings

Relationships

- <<include>>
 - Book Ticket → Make Online Payment
- <<extend>>
 - Cancel Booking → Book Ticket
- Generalization
 - Theater Manager and Administrator are specialized types of System User.

DIAGRAM:



Explanation :

- Customer searches movies, selects seats, books or cancels tickets, and makes online payments.
- Theater Manager updates movie schedules and seat availability.
- Administrator manages user accounts and system settings.
- Payment Gateway processes online payments.
- The Book Ticket use case includes Make Online Payment because payment is mandatory for booking.
- Cancel Booking extends Book Ticket because cancellation occurs only after a ticket has been booked.
- Theater Manager and Administrator are specialized users of the system through generalization.

2. Classes and Attributes

1. Student

- StudentID
- Name
- Email
- Phone

Methods

- registerCourse()
- viewAttendance()
- viewResult()

2. Faculty

- FacultyID
- Name
- Designation

Methods

- teachCourse()
- markAttendance()
- uploadMarks()

3. Department

- DepartmentID
- DepartmentName

Methods

- addCourse()
- assignFaculty()

4. Course

- CourseID
- CourseName
- Credits

Methods

- assignStudent()
- scheduleClass()

5. Attendance

- AttendanceID
- Date
- Status

Methods

- markAttendance()
- generateReport()

6. Examination

- ExamID
- ExamName

- Marks

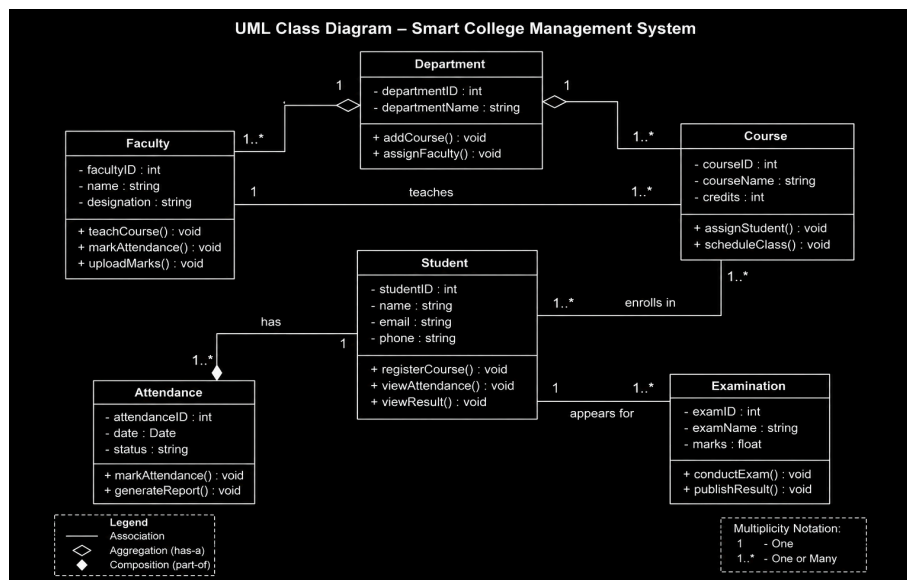
Methods

- conductExam()
- publishResult()

Relationships

- Department contains Faculty and Courses (Aggregation)
- Student enrolls in Course (Association)
- Faculty teaches Course (Association)
- Student has Attendance (Composition)
- Student appears for Examination (Association)

Diagram:



3. Objects (Participants)

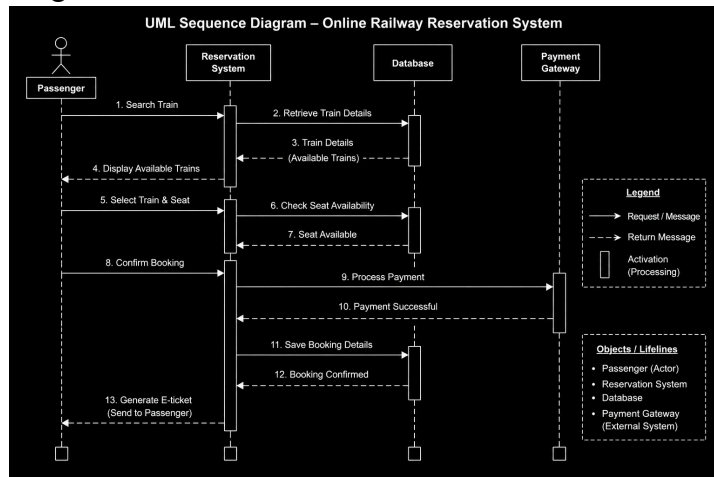
1. Passenger
2. Reservation System
3. Database
4. Payment Gateway

Sequence of Operations

1. Passenger → Reservation System : Search Train
2. Reservation System → Database : Retrieve Train Details
3. Database → Reservation System : Display Available Trains
4. Passenger → Reservation System : Select Train & Seat
5. Reservation System → Database : Check Seat Availability
6. Database → Reservation System : Seat Available
7. Passenger → Reservation System : Confirm Booking
8. Reservation System → Payment Gateway : Process Payment

9. Payment Gateway → Reservation System : Payment Successful
10. Reservation System → Database : Save Booking Details
11. Database → Reservation System : Booking Confirmed
12. Reservation System → Passenger : Generate & Send E-Ticket

Diagram:



Explanation:

- The Passenger searches for available trains.
- The Reservation System retrieves train information from the Database.
- After the passenger selects a train and seat, the system checks seat availability.
- If seats are available, the passenger confirms the booking.
- The system processes the payment through the Payment Gateway.
- Once payment is successful, booking details are stored in the database.
- Finally, the system generates and sends the e-ticket to the passenger.

4. Activities

1. Start
2. Submit Claim
3. Upload Required Documents
4. Verify Documents
5. Validate Claim Details
6. Decision: Is the claim valid?
 - Yes → Approve Claim
 - Process Compensation
 - Send Confirmation
 - End
 - No → Reject Claim
 - Notify Customer
 - End

Parallel Activities

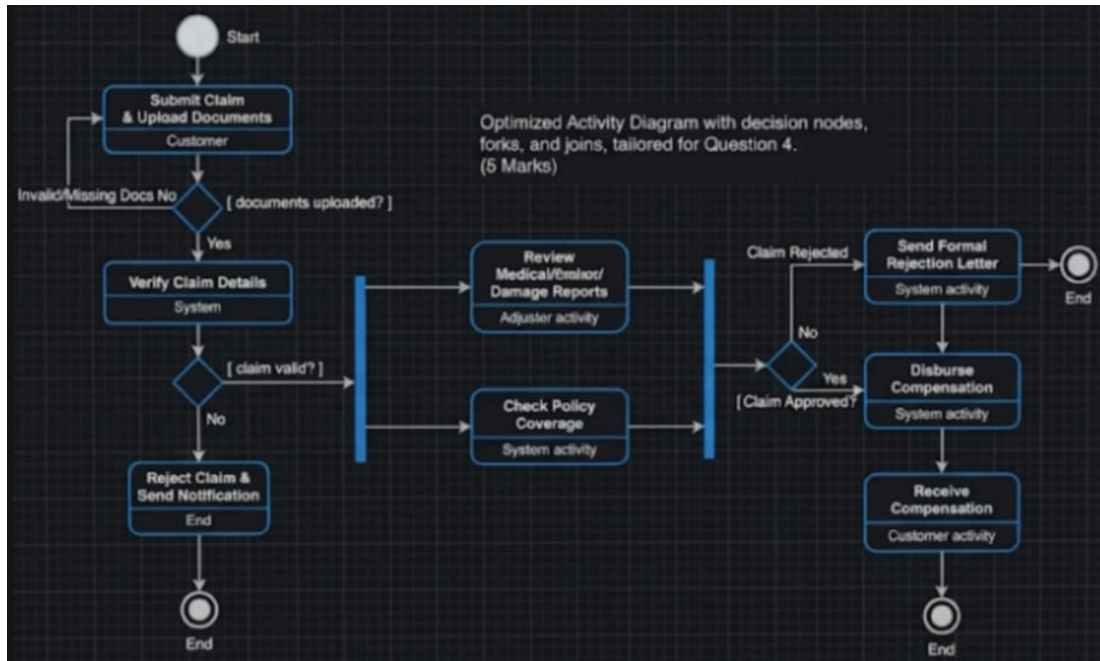
After submitting the claim:

- Verify Documents

- Validate Claim Details

These two activities occur simultaneously before the decision is made.

Diagram:



Explanation

The Activity Diagram illustrates the workflow of an insurance claim processing system.

- The process begins when the customer submits a claim.
- The required documents are uploaded for verification.
- The system performs document verification and claim validation simultaneously.
- A decision node checks whether the claim is valid.
- If the claim is valid, it is approved, compensation is processed, and a confirmation is sent to the customer.
- If the claim is invalid, the claim is rejected and the customer is notified.
- The process then ends.

5. States

1. Initial State (Start)
2. Idle
3. Water Filling
4. Washing
5. Rinsing
6. Spinning
7. Completed
8. Final State (End)

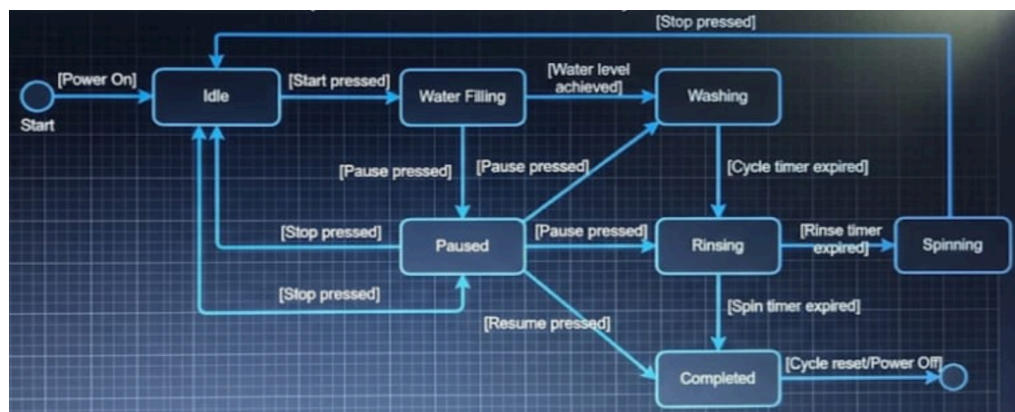
State Transitions

- Start → Idle
- Idle → Water Filling
- Water Filling → Washing
- Washing → Rinsing
- Rinsing → Spinning
- Spinning → Completed
- Completed → End

Additional States

- Pause: Machine operation is temporarily stopped.
- Resume: Machine continues from the paused state to the previous operation (Washing, Rinsing, or Spinning).

Diagram:



Explanation

The State Diagram represents the different states of an automatic washing machine during its operation.

- Initially, the machine is in the Idle state waiting for the user to start a wash cycle.
- It then moves to Water Filling, followed by Washing, Rinsing, and Spinning.
- After all washing operations are completed, the machine enters the Completed state and then reaches the Final State.
- The machine also supports Pause and Resume functionality, allowing the washing cycle to be temporarily stopped and continued without restarting the process.

6. Hardware Nodes

1. Smart Meter
2. Edge Gateway
3. Cloud Server
4. Utility Dashboard (Desktop/Web)
5. Mobile Device

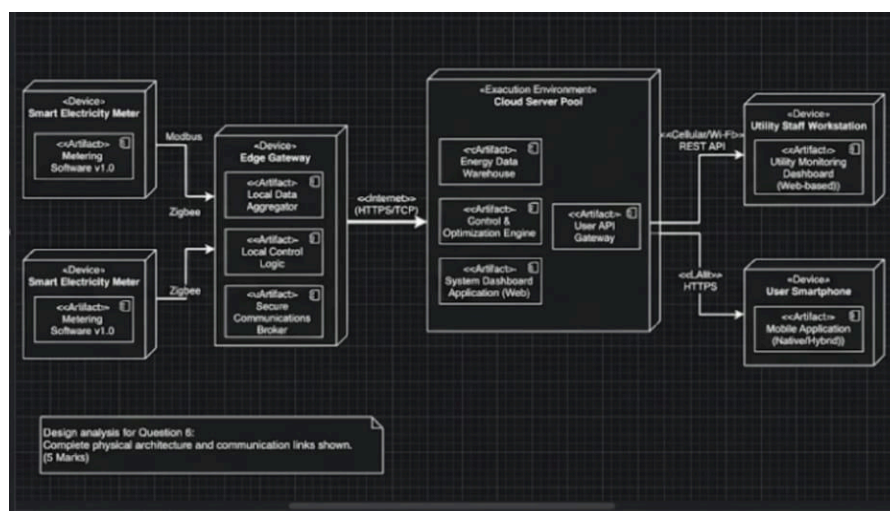
Software Artifacts

- Energy Monitoring Application
- Database
- Analytics Module
- Control Service

Communication

- Smart Meter → Edge Gateway
- Edge Gateway → Cloud Server
- Cloud Server → Utility Dashboard
- Cloud Server → Mobile Device

Diagram:



Explanation:

The Deployment Diagram illustrates the physical architecture of the Smart Energy Monitoring System.

- The Smart Meter collects electricity consumption data from households.
- The collected data is transmitted to the Edge Gateway, which acts as an intermediate communication device.
- The Cloud Server stores the data in a Database, runs the Analytics Module, and manages the Control Service.
- The processed information is then made available through the Utility Dashboard for administrators and the Mobile Device application for users to monitor energy usage in real time.
- The communication links between these hardware nodes represent the deployment and interaction of software components across the system.